Follow-up 1 (Bethesda category deltas, surgical cohorts, post-2019) Identify surgical cohort studies (2019–2025) that report risk of malignancy (ROM) by Bethesda category with and without NIFTP counted as malignant, and provide absolute ROM deltas for Bethesda III, IV, and V. Extract denominators and how ROM is defined (resected nodules vs resected FNAs vs all FNAs). Prefer studies that specify whether 2017 vs 2018 NIFTP criteria were applied.

Surgical cohort studies from 2019-2025 report highly variable absolute ROM deltas following NIFTP reclassification—ranging from 0-20% for Bethesda III, 0.7-30.8% for Bethesda IV, and 0-41.5% for Bethesda V—with most using resected nodules as the denominator, though only one study explicitly applied the 2018 NIFTP criteria (showing minimal impact), indicating that reported ROM changes depend heavily on geographic population, institutional resection practices, and which diagnostic criteria were used.

Abstract

This systematic review identified 25 sources examining the impact of NIFTP reclassification on risk of malignancy (ROM) across Bethesda categories, though only a subset met the strict criteria of surgical cohort studies from 2019-2025 with complete ROM delta reporting. Most studies calculated ROM using resected nodules or resected FNAs as the denominator , with fewer using all FNAs . Critically, only three studies specified which NIFTP diagnostic criteria were applied: J. Kopczyński et al. (2020) used the revised 2018 criteria , Milim Kim et al. (2018) used 2017 criteria , and A. Bychkov et al. (2018) applied the original 2016 Nikiforov criteria . For Bethesda III (AUS/FLUS), absolute ROM deltas ranged from 0% to 20% . For Bethesda IV (FN/SFN), deltas ranged from 0.7% to 30.8% . For Bethesda V (suspicious for malignancy), deltas ranged from 0% to 41.5% .

The evidence demonstrates substantial heterogeneity explained primarily by geographic region and NIFTP diagnostic criteria applied. Western cohorts consistently showed larger ROM decreases than Asian cohorts, where NIFTP prevalence was markedly lower (0.4-2.9% vs. 7-25% of relevant cases) . The study using 2018 criteria found only 5 NIFTP cases and minimal ROM impact (0.7% for Bethesda IV, 4.3% for Bethesda V) , suggesting stricter criteria substantially reduce case identification. These findings support recommendations that institutions establish their own NIFTP rates and ROM adjustments rather than applying universal values , and highlight that studies failing to specify NIFTP criteria version may not be directly comparable.

Paper search

We performed a semantic search using the query "Follow-up 1 (Bethesda category deltas, surgical cohorts, post-2019)

Identify surgical cohort studies (2019–2025) that report risk of malignancy (ROM) by Bethesda category with and without NIFTP counted as malignant, and provide absolute ROM deltas for Bethesda III, IV, and V. Extract denominators and how ROM is defined (resected nodules vs resected FNAs vs all FNAs). Prefer studies that specify whether 2017 vs 2018 NIFTP criteria were applied." across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We retrieved the 500 papers most relevant to the query.

Screening

We screened in sources based on their abstracts that met these criteria:

- **Surgical Cohort with Histopathology**: Does this study report on patients who underwent thyroid surgery/resection with histopathological examination of resected specimens to determine final malignancy status?
- **Bethesda System Classification**: Does this study report thyroid cytology results using the Bethesda System for Reporting Thyroid Cytopathology?
- **Stratified ROM Data**: Does this study provide risk of malignancy (ROM) data stratified by individual Bethesda categories, particularly categories III (AUS/FLUS), IV (FN/SFN), and/or V (SM)?
- **NIFTP Consideration**: Does this study address NIFTP (Noninvasive Follicular Thyroid Neoplasm with Papillary-like Nuclear Features) or provide data that allows calculation of ROM both with and without NIFTP counted as malignant?
- **Clear Denominator Reporting**: Does this study clearly specify what denominator is used for ROM calculations (e.g., resected nodules, resected FNAs, or all FNAs)?
- **Adequate Sample Size**: Does this study include an adequate sample size (≥20 cases) rather than being a case report or small case series?
- **Adult Population**: Does this study include adult patients (not exclusively pediatric populations)?
- **Full Manuscript**: Is this a full manuscript (not a conference abstract, letter, or editorial) with sufficient methodological detail and data?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Study Design**:

  Extract study design and timeframe details including:

- Study type (confirm it's a surgical cohort study)
- Study period (verify it's 2019-2025 or overlaps with this period)
- Institution type (single vs multi-institutional)
- Total number of patients/procedures included

- **NIFTP Criteria**:

  Extract information about NIFTP diagnostic criteria used:

- Whether 2017 or 2018 NIFTP criteria were specified
- Who performed the NIFTP review/reclassification (pathologists, study team)
- Whether cases were re-reviewed specifically for NIFTP or identified prospectively
- Any mention of specific NIFTP diagnostic features applied

- **ROM Definition**:

  Extract how risk of malignancy (ROM) was defined and calculated:

- Denominator used (all FNAs, resected FNAs only, or resected nodules only)

- Whether ROM calculation included all Bethesda categories or specific ones
- How surgical follow-up was determined
- Time period between FNA and surgery if specified

- **NIFTP Cases**:

  Extract NIFTP case details including:

- Total number of NIFTP cases identified
- NIFTP prevalence (as percentage of resected cases, malignant cases, or total cohort)
- Distribution of NIFTP cases across Bethesda categories III, IV, V
- Any demographic or clinical characteristics of NIFTP cases if reported

- **Bethesda III ROM**:

  Extract ROM data for Bethesda III (AUS/FLUS) including:

- ROM when NIFTP counted as malignant (percentage and numerator/denominator)
- ROM when NIFTP counted as non-malignant (percentage and numerator/denominator)
- Absolute ROM delta (absolute percentage point change)
- Relative ROM change if reported

- **Bethesda IV ROM**:

  Extract ROM data for Bethesda IV (FN/SFN) including:

- ROM when NIFTP counted as malignant (percentage and numerator/denominator)
- ROM when NIFTP counted as non-malignant (percentage and numerator/denominator)
- Absolute ROM delta (absolute percentage point change)
- Relative ROM change if reported

- **Bethesda V ROM**:

  Extract ROM data for Bethesda V (suspicious for malignancy) including:

- ROM when NIFTP counted as malignant (percentage and numerator/denominator)
- ROM when NIFTP counted as non-malignant (percentage and numerator/denominator)
- Absolute ROM delta (absolute percentage point change)
- Relative ROM change if reported

- **Treatment Impact**:

  Extract information about treatment implications of NIFTP reclassification:

- Extent of surgery performed in NIFTP cases (lobectomy vs total thyroidectomy)
- Rate of completion thyroidectomy in NIFTP cases
- Use of radioactive iodine treatment in NIFTP cases
- Any mention of overtreatment or treatment recommendations changes

Characteristics of Included Studies

This systematic review identified 25 sources examining the impact of noninvasive follicular thyroid neoplasm with papillary-like nuclear features (NIFTP) reclassification on risk of malignancy (ROM) across Bethesda categories. The sources comprised primary surgical cohort studies as well as systematic reviews and meta-analyses that aggregated data from multiple institutions.

Study	Full text retrieved?	Study Type	Study Period	Institution Type	Sample Size	NIFTP Criteria	ROM Denominator
Samantha M. Linhares et al., 2020	No	Retrospective surgical cohort	Not mentioned	Single institution	565 patients	Not specified	Resected FNAs
Mara Ventura et al., 2019	No	Surgical cohort	2012-2014	Not mentioned	772 patients	Not specified	All FNAs
J. Kopczyński et al., 2019	No	Surgical cohort	Not mentioned	Not mentioned	998 nodules	Not specified	Resected nodules
C. Rana et al., 2021	No	Retrospective surgical cohort	Not mentioned	Not mentioned	Not mentioned	Not specified	Resected nodules
M. Bongiovanni et al., 2019	No	Systematic review and meta-analysis	Literature search Nov 2018	Multi-institutional	13,752 thyroidectomies	Not specified	Resected nodules
Simon Sung et al., 2019	No	Surgical cohort	Jan 2013-Aug 2017	Single institution	4,500 FNA cases	Not specified	Resected FNAs
H. Vuong et al., 2019	No	Meta-analysis	Jan 2016-Jul 2018	Multi-institutional	14,153 resected nodules	Not specified	Resected nodules
Melissa L. Mao et al., 2018	No	Surgical cohort	Jan 2010-Apr 2016	Single institution	847 patients	Not specified	Resected nodules
Brenessa Lindeman et al., 2018	No	Retrospective cohort	Apr 2016-Feb 2017	Single institution	353 patients	Not specified	Resected nodules
R. Rosenblum et al., 2019	No	Retrospective study	Jan 2013-Dec 2015	Single institution	2,674 patients	Not specified	All FNAs
J. Kopczyński et al., 2020	No	Surgical cohort	Not mentioned	Single institution	998 nodules	2018 criteria	Resected nodules
Ryan P. Lau et al., 2017	No	Surgical cohort	Jan 2013-Jun 2016	Single institution	750 FNAs	Not specified	Resected FNAs
K. Strickland et al., 2015	No	Surgical cohort	22-month period	Not mentioned	655 FNAs	Encapsulation criteria specified	Resected FNAs

Study	Full text retrieved?	Study Type	Study Period	Institution Type	Sample Size	NIFTP Criteria	ROM Denominator
H. Al-Maghrabi et al., 2022	No	Retrospective cohort	Jan 2013-Dec 2017	Single institution	1,066 FNAC specimens	Not specified	Resected nodules
Milim Kim et al., 2018	Yes	Surgical cohort	2012-2014	Single institution	5,549 FNAC cases	2017 criteria	Resected cases and all FNAC
Bryan Wei Wen Lee et al., 2021	No	Surgical cohort	Jan 2010-Aug 2016	Single institution	788 patients	Not specified	Resected nodules
E. Haaga et al., 2021	No	Systematic review and meta-analysis	Mar-Jun 2020	Multi-institutional	2,553 NIFTP cases	Not specified	Surgically resected cases
A. Bychkov et al., 2018	Yes	Multi-institutional surgical cohort	2010-2017	Multi-institutional (6 centers, 5 Asian countries)	11,372 nodules	2016 Nikiforov criteria	Resected nodules
Prerna Guleria et al., 2019	No	Retrospective study	4.5-year period	Single institution	2,554 aspirates	Not specified	Resected FNAs
C. Kiernan et al., 2018	No	Surgical cohort	Not mentioned	Single institution	1,046 patients	Not specified	Resected FNAs
H. Y. Na et al., 2019	No	Surgical cohort (CNB)	Not mentioned	Not specified	2,687 CNBs	Not specified	Resected nodules
A. Delman et al., 2022	No	Surgical cohort	2016-2019	Multi-institutional (NSQIP)	13,121 patients	Not mentioned	Resected nodules
Yan-li Zhu et al., 2020	Yes	Retrospective surgical cohort	Apr 2014-Mar 2019	Single institution	2,781 FNAs	Not specified	Resected FNAs
N. Hacım et al., 2021	Yes	Retrospective surgical cohort	Approximately 2020	Single institution	159 patients	Not mentioned	Resected nodules
Daniel N. Johnson et al., 2018	No	Surgical cohort	Not mentioned	Not mentioned	1,396 nodules	Not specified	Resected nodules

The majority of included studies were single-institution retrospective surgical cohort studies. Only three studies explicitly specified which NIFTP diagnostic criteria were applied: J. Kopczyński et al. (2020) used the revised 2018 criteria , Milim Kim et al. (2018) used the 2017 criteria , and A. Bychkov et al. (2018) applied the original 2016 Nikiforov criteria . Most studies calculated ROM using resected nodules or resected FNAs as the denominator, though some studies including Mara Ventura et al. (2019) and R. Rosenblum et al. (2019) used all FNAs .

NIFTP Prevalence and Distribution

Study	Total NIFTP Cases	NIFTP Prevalence	Bethesda III Distribution	Bethesda IV Distribution	Bethesda V Distribution
Mara Ventura et al., 2019	15	Not mentioned	10 cases	2 cases	3 cases
C. Rana et al., 2021	Not mentioned	6.9% of excised nodules, 16.8% of neoplastic lesions	Not mentioned	Not mentioned	Not mentioned
M. Bongiovanni et al., 2019	Not mentioned	14% for Bethesda V, 3% for Bethesda VI	Not mentioned	Not mentioned	14%
Simon Sung et al., 2019	36	13% of malignant neoplasms	14 cases	13 cases	Not mentioned
Melissa L. Mao et al., 2018	32	22% of FVPTC subgroup	Not mentioned	Not mentioned	Not mentioned
Brenessa Lindeman et al., 2018	26	7.3% of total cohort	13 (50%)	4 (15%)	6 (23%)
R. Rosenblum et al., 2019	Not mentioned	9.5% of malignant tumors	Not mentioned	Not mentioned	Not mentioned
J. Kopczyński et al., 2020	5	2.3% of PTC diagnoses	Not mentioned	1 case	3 cases
Ryan P. Lau et al., 2017	87	Not mentioned	Not mentioned	Not mentioned	Not mentioned
K. Strickland et al., 2015	85	24.6% of malignancies	Not mentioned	Not mentioned	Not mentioned
Milim Kim et al., 2018	25	1.5% of PTCs	14 cases (AUS)	2 cases	3 cases
Bryan Wei Wen Lee et al., 2021	10	1.2%	Not mentioned	Not mentioned	Not mentioned
E. Haaga et al., 2021	2,553	4.4% of resected cases	29.2%	24.2%	19.5%
A. Bychkov et al., 2018	59	2.9% of excised nodules, 5.3% of malignant	22.0%	32.2%	11.9%
Prerna Guleria et al., 2019	13	Not mentioned	13 cases (AUS subcategories)	Not mentioned	Not mentioned
C. Kiernan et al., 2018	17	46% of evaluated cases	Not mentioned	Not mentioned	Not mentioned

Study	Total NIFTP Cases	NIFTP Prevalence	Bethesda III Distribution	Bethesda IV Distribution	Bethesda V Distribution
H. Y. Na et al., 2019	32	4.7% of PTC cases	11 (indeterminate, 34.4%)	20 (FN, 62.5%)	1 (3.1%)
Yan-li Zhu et al., 2020	4	0.4% of PTC cases	Not mentioned	1 case	Not mentioned

The meta-analysis by E. Haaga et al. (2021) pooled data from 58 studies and found that NIFTP comprised 4.4% of all surgically resected cases . The majority of NIFTP cases (79.0%) fell within the indeterminate Bethesda categories . Distribution across categories showed 29.2% in AUS/FLUS (Bethesda III), 24.2% in follicular neoplasm (Bethesda IV), and 19.5% in suspicious for malignancy (Bethesda V) . Notably, NIFTP prevalence varied considerably by geographic region, with Western studies reporting higher rates than Asian cohorts. A. Bychkov et al. (2018) found that Asian countries had lower NIFTP rates (2.9% of excised nodules) compared to Western reports , and H. Vuong et al. (2019) confirmed that decreases in ROM were more prominent in Western than Asian cohorts .

Risk of Malignancy by Bethesda Category

Bethesda III (AUS/FLUS)

Study	ROM with NIFTP Malignant	ROM with NIFTP Non-malignant	Absolute Delta	Relative Change
Samantha M. Linhares et al., 2020	55%	48%	7%	Not mentioned
Mara Ventura et al., 2019	Not mentioned	Not mentioned	5.5%	15.2%
C. Rana et al., 2021	40.0%	20.0%	20%	Not mentioned
Simon Sung et al., 2019	Not mentioned	Not mentioned	15%	Not mentioned
H. Vuong et al., 2019	Not mentioned	Not mentioned	Not mentioned	32% (95% CI: 24-39)
Melissa L. Mao et al., 2018	Not mentioned	Not mentioned	8%	Not mentioned
Brenessa Lindeman et al., 2018	33%	18%	15%	Not mentioned
R. Rosenblum et al., 2019	14%	12.5%	1.5%	Not mentioned
Ryan P. Lau et al., 2017	42.3%	22.3%	20%	Not mentioned
K. Strickland et al., 2015	39.2%	21.6%	17.6%	45%
H. Al-Maghrabi et al., 2022	50%	39.2%	10.8%	Not mentioned

Study	ROM with NIFTP Malignant	ROM with NIFTP Non-malignant	Absolute Delta	Relative Change
Milim Kim et al., 2018	75.8%	71.2%	4.6%	6.1%
Bryan Wei Wen Lee et al., 2021	28.1%	26.3%	1.8%	Not mentioned
A. Bychkov et al., 2018	44.7%	38.6%	6.0%	13.5%
Prerna Guleria et al., 2019	51.8%	35.8%	16%	31%
C. Kiernan et al., 2018	17%	15%	2%	Not mentioned
Yan-li Zhu et al., 2020	50.0%	50.0%	0%	Not mentioned

For Bethesda III, absolute ROM deltas ranged widely from 0% to 20%. The meta-analysis by H. Vuong et al. (2019) found category III had the largest relative decrease in ROM at 32% . Western institutions generally showed larger absolute decreases, while Asian studies reported minimal impact. Yan-li Zhu et al. (2020) from China found no change in ROM (0% delta) , and Bryan Wei Wen Lee et al. (2021) from Singapore reported only a 1.8% absolute decrease . Milim Kim et al. (2018) from South Korea similarly found a modest 4.6% absolute decrease . In contrast, C. Rana et al. (2021) from India reported a 20% absolute decrease , and Ryan P. Lau et al. (2017) found a 20% decrease at their Western academic institution .

Bethesda IV (FN/SFN)

Study	ROM with NIFTP Malignant	ROM with NIFTP Non-malignant	Absolute Delta	Relative Change
Samantha M. Linhares et al., 2020	50%	35%	15%	Not mentioned
Mara Ventura et al., 2019	Not mentioned	Not mentioned	1.8%	7.6%
C. Rana et al., 2021	46.5%	27.4%	19.1%	Not mentioned
Simon Sung et al., 2019	Not mentioned	Not mentioned	16.2%	Not mentioned
Melissa L. Mao et al., 2018	Not mentioned	Not mentioned	10%	Not mentioned
Brenessa Lindeman et al., 2018	Not mentioned	22%	7%	Not mentioned
R. Rosenblum et al., 2019	26.7%	25%	1.7%	Not mentioned
J. Kopczyński et al., 2020	Not mentioned	Not mentioned	0.7%	Not mentioned
Ryan P. Lau et al., 2017	48.7%	17.9%	30.8%	Not mentioned

Study	ROM with NIFTP Malignant	ROM with NIFTP Non-malignant	Absolute Delta	Relative Change
K. Strickland et al., 2015	45.5%	37.5%	8%	18%
H. Al-Maghrabi et al., 2022	53.8%	33.3%	20.5%	Not mentioned
Bryan Wei Wen Lee et al., 2021	26.7%	20.0%	6.7%	Not mentioned
A. Bychkov et al., 2018	40.4% (78/193)	30.6% (59/193)	9.8%	24.4%
C. Kiernan et al., 2018	23%	21%	2%	Not mentioned
Yan-li Zhu et al., 2020	34.5%	31.0%	3.5%	Not mentioned

Bethesda IV showed the greatest relative reduction in ROM according to A. Bychkov et al. (2018), with a 24.4% relative decrease . Ryan P. Lau et al. (2017) reported the largest absolute decrease at 30.8 percentage points . The study by J. Kopczyński et al. (2020), which used the revised 2018 NIFTP criteria, found only a 0.7% decrease in ROM for this category , suggesting that the stricter 2018 criteria substantially reduced the number of cases qualifying as NIFTP. H. Al-Maghrabi et al. (2022) from Saudi Arabia reported a statistically significant decrease in ROM for the FN/SFN category from 53.8% to 33.3% .

Bethesda V (Suspicious for Malignancy)

Study	ROM with NIFTP Malignant	ROM with NIFTP Non-malignant	Absolute Delta	Relative Change
Samantha M. Linhares et al., 2020	93%	91%	2%	Not mentioned
Mara Ventura et al., 2019	Not mentioned	Not mentioned	10.3%	14.2%
C. Rana et al., 2021	88.8%	66.3%	22.5%	Not mentioned
M. Bongiovanni et al., 2019	Not mentioned	Not mentioned	14%	Not mentioned
H. Vuong et al., 2019	Not mentioned	Not mentioned	16%	Not mentioned
Melissa L. Mao et al., 2018	Not mentioned	Not mentioned	3%	Not mentioned
Brenessa Lindeman et al., 2018	91%	73%	18%	~19.8%
J. Kopczyński et al., 2020	Not mentioned	Not mentioned	4.3%	Not mentioned
Ryan P. Lau et al., 2017	93.6%	61.7%	31.9%	Not mentioned
K. Strickland et al., 2015	87.2%	45.7%	41.5%	~50%

Study	ROM with NIFTP Malignant	ROM with NIFTP Non-malignant	Absolute Delta	Relative Change
H. Al-Maghrabi et al., 2022	85%	75%	10%	Not mentioned
Milim Kim et al., 2018	99.2% (390/393)	98.5% (387/393)	0.7%	Not mentioned
Bryan Wei Wen Lee et al., 2021	89.2%	87.7%	1.5%	Not mentioned
A. Bychkov et al., 2018	88.0% (125/142)	83.1% (118/142)	4.9%	5.6%
C. Kiernan et al., 2018	68%	60%	8%	Not mentioned
Yan-li Zhu et al., 2020	78.5%	78.5%	0%	Not mentioned

The meta-analysis by H. Vuong et al. (2019) found that Bethesda V had the largest absolute decrease in ROM at 16 percentage points . K. Strickland et al. (2015) reported a remarkable 41.5 percentage point decrease with a nearly 50% relative reduction . However, Asian cohorts showed markedly different results. Milim Kim et al. (2018) found only a 0.7% absolute decrease , Bryan Wei Wen Lee et al. (2021) reported 1.5% , and Yan-li Zhu et al. (2020) found no change (0%) . This geographic disparity is consistent with the lower NIFTP prevalence observed in Asian populations.

Treatment Implications

Study	Extent of Surgery	Completion Thyroidectomy Rate	Radioactive Iodine Use	Overtreatment Noted
Mara Ventura et al., 2019	93% total thyroidectomy	Not mentioned	20%	Not mentioned
C. Rana et al., 2021	Lobectomy preferred	Not mentioned	Not mentioned	16.2% overtreatment
M. Bongiovanni et al., 2019	Not mentioned	Not mentioned	Not mentioned	Yes, avoid overtreatment emphasized
Melissa L. Mao et al., 2018	16 total thyroidectomy, 16 lobectomy	75% (12/16)	28% (9 patients)	NIFTP is indolent, no further surgery needed
Brenessa Lindeman et al., 2018	17 lobectomy, 9 total thyroidectomy	Not mentioned	Not mentioned	89% could have had lobectomy only
Milim Kim et al., 2018	Less extensive surgery implied	Lower rate implied	Reduced use implied	Conservative management recommended

Study	Extent of Surgery	Completion Thyroidectomy Rate	Radioactive Iodine Use	Overtreatment Noted
A. Bychkov et al., 2018	Lobectomy preferred over total thyroidectomy	Not mentioned	Not mentioned	Reduced overtreatment with less aggressive treatment
C. Kiernan et al., 2018	Not mentioned	Not mentioned	Not mentioned	NIFTP helps avoid excessive treatment

Treatment data revealed substantial overtreatment prior to NIFTP recognition. Melissa L. Mao et al. (2018) found that among NIFTP patients, half underwent total thyroidectomy and 75% of those who initially had lobectomy went on to have completion thyroidectomy . Additionally, 28% received radioactive iodine treatment . Brenessa Lindeman et al. (2018) determined that 89% of patients who underwent total thyroidectomy could have been definitively treated with lobectomy alone . C. Rana et al. (2021) reported that inability to diagnose NIFTP preoperatively led to overtreatment in 16.2% of cases and recommended diagnostic lobectomy over total thyroidectomy .

Synthesis

The data reveal substantial heterogeneity in ROM changes across studies, which can be explained by several key factors.

Geographic and Population Differences

The most striking pattern is the geographic disparity between Western and Asian cohorts. H. Vuong et al. (2019) explicitly noted that ROM decreases were more prominent in Western than Asian practice . This finding is mechanistically explained by the lower prevalence of NIFTP in Asian populations—Yan-li Zhu et al. (2020) found NIFTP accounted for only 0.4% of papillary thyroid carcinomas , compared to 13% of malignant neoplasms in Simon Sung et al. (2019) and 24.6% of malignancies in K. Strickland et al. (2015) . This prevalence difference directly determines the magnitude of ROM change: institutions with low NIFTP prevalence will necessarily see smaller absolute ROM deltas regardless of baseline ROM values.

Impact of NIFTP Diagnostic Criteria

The evolution from 2016 to 2018 diagnostic criteria substantially affects case identification. J. Kopczyński et al. (2020), the only study explicitly using revised 2018 criteria, found only 5 NIFTP cases (2.3% of papillary carcinomas) with minimal ROM decreases (0.7% for Bethesda IV, 4.3% for Bethesda V) . In contrast, studies using earlier criteria reported much higher NIFTP prevalence and larger ROM changes. This suggests the stricter 2018 criteria, which added requirements to avoid misdiagnosis of papillary thyroid cancer as NIFTP , substantially narrow the population of tumors qualifying for reclassification.

Institutional Resection Practices

H. Vuong et al. (2019) identified a positive correlation between NIFTP rate and surgical resection rate (r=0.83, P=0.02) . Institutions with more aggressive surgical management of indeterminate nodules will necessarily identify more NIFTP cases, as these tumors cannot be diagnosed without histological examination. This creates apparent variation in NIFTP impact that reflects institutional practice patterns rather than true biological differences.

ROM Denominator Definitions

Studies using all FNAs as the denominator (Mara Ventura et al., 2019 ; R. Rosenblum et al., 2019) will show different absolute ROM values than those using only resected nodules, as the former includes cases that never received surgical confirmation. However, when calculating the delta between ROM with and without NIFTP as malignant, this methodological difference has less impact since the same denominator is used in both calculations.

Reconciliation of Findings

The data suggest that NIFTP reclassification has meaningful clinical impact primarily in Western populations and institutions with higher surgical resection rates for indeterminate nodules. For Bethesda III, Western studies show absolute decreases of 7-20 percentage points , while Asian studies report 0-6 percentage points . For Bethesda IV, the pattern is similar with Western decreases of 7-31 percentage points versus Asian decreases of 0.7-9.8 percentage points . Bethesda V shows the widest variability, with some Western studies reporting dramatic decreases up to 41.5 percentage points while Asian studies consistently show minimal impact (0-4.9 percentage points) .

These findings support the recommendation by multiple authors that institutions should establish their own NIFTP rates and corresponding ROM adjustments rather than applying universal values . The 2018 NIFTP criteria appear to substantially reduce case identification, and studies applying these stricter criteria will show markedly lower impact on ROM. Treatment data consistently demonstrate that NIFTP reclassification reduces overtreatment, with studies suggesting 16-89% of NIFTP patients received more aggressive treatment than necessary .

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